Special Notice

Area-Multiplied Photonic-crystal Enhanced Devices (AMPED)
Proposers Day
DARPA-SN-26-63
April 8, 2026

Defense Advanced Research Projects Agency
Microsystems Technology Office
675 North Randolph Street
Arlington, VA 22203-2114

AMPED Proposers Day
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Area-Multiplied Photonic-crystal Enhanced Devices (AMPED)
Defense Advanced Research Projects Agency (DARPA)
Microsystems Technology Office (MTO)

Posting Date: April 8, 2026
Event Date: May 13, 2026, at 8:00 a.m. Eastern Time (ET)
Registration Deadline: April 29, 2026, at 5:00 p.m. Eastern Time (ET)
Registration Website: https://cvent.me/Veexab
Attendance is restricted to U.S. Persons only (U.S. Citizens and Permanent Residents)
Technical POC: Dr. David Meyer, MTO Program Manager
E-mail: AMPED@darpa.mil

PROPOSERS DAY DESCRIPTION:

The Defense Advanced Research Projects Agency (DARPA) will host an in-person Proposers Day in support of the DARPA Program Solicitation (PS) for AMPED on May 13, 2026, at the DARPA Conference Center (DCC), Arlington, VA, from 8:00 a.m. to 4:00 p.m. ET. DARPA anticipates releasing the AMPED Program Solicitation (PS) prior to Proposers Day. If released, the PS will be made available at https://sam.gov/.

The goals of the Proposers Day are:

(1) To introduce the science and technology community (industry, academia, and government) to the AMPED program vision and goals;
(2) To facilitate interaction between researchers with capabilities and interests relevant to the AMPED program goals;
(3) To collect questions during the event that DARPA may answer and make available in a Question and Answer (Q&A) document; and
(4) To encourage and promote teaming arrangements among organizations that have the relevant expertise, research facilities, and capabilities for executing research and development responsive to the AMPED program goals.

The Proposers Day will include overview presentations by government personnel and opportunities for team building among the participants. The meeting will start with presentations by DARPA, including a technical briefing and a contracting discussion to further describe the planned PS objectives and content. These will be followed by a Q&A session, a poster session, and breakout meetings with the Program Manager.

PROGRAM OBJECTIVE AND DESCRIPTION:

The Area-Multiplied Photonic-crystal Enhanced Devices (AMPED) program aims to create a new paradigm in scalable semiconductor laser technology. By leveraging novel photonic structures, AMPED will overcome fundamental scaling limitations of current laser designs, creating the foundation for a highly manufacturable, low-size-weight-and-power (SWaP), and widely proliferated laser technology for the Department of War (DoW) and relevant commercial applications. AMPED will demonstrate single-mode, diffraction-limited, continuous-wave laser devices capable of high power.

The ability to increase the power of today's single-mode semiconductor lasers is fundamentally constrained by physical limitations inherent to their traditional device architecture. This constraint manifests as two primary technical challenges:

Technical Challenge 1: Area scaling for single-mode emission

Single-mode semiconductor lasers such as edge-emitting lasers (EELs) and vertical-cavity surface-emitting lasers (VCSELs) are fundamentally limited to 1D scaling. While their output power generally scales with the volume of the active gain region, expanding this region in the transverse directions allows for unwanted higher-order lasing modes to emerge, which significantly degrades the beam quality. In other words, the physical volume overlap of the cavity geometry and the gain medium geometry prevents independent scaling of each. Photonic-crystal surface-emitting lasers (PCSELs) offer a path forward by physically separating the gain medium from the optical cavity, but current PCSEL designs face their own scaling challenges. As the PCSEL device diameter increases, the mode margin between the desired fundamental mode gain and competing higher-order mode gain collapses rapidly, making it very difficult to maintain a stable, single-mode beam beyond a certain size. Current approaches using simple photonic crystal lattice designs are approaching their single-mode limits and cannot deliver further scaling.

Technical Challenge 2: Maximizing power density by minimizing loss

Once laser device area is maximized, power can be further increased by increasing the number of photons generated per unit area. However, this power conversion efficiency is reduced by optical, electrical, and other losses. Stacking multiple gain regions vertically, a successful technique used in VCSELs, may be limited in current PCSEL designs because n++/p++ tunnel junctions between gain layers introduce substantial free-carrier absorption, counteracting the benefits of additional gain. Similarly, as more features are designed into the photonic crystal to extend single-mode operation to larger device size, the area of high conductivity material shrinks leading to more resistive loss in the vertical path of current flow.

Today, these challenges are handled with incremental modifications to existing PCSEL designs, which have yielded diminishing returns. It has become apparent that the search for piecewise improvements cannot solve the problem of limited power scaling. The key to addressing these two technical challenges is to redirect the effort to a fundamental redesign of the laser device's architecture, leveraging all three dimensions of design flexibility. AMPED seeks innovative strategies that combine novel in-plane photonic crystal designs with out-of-plane epitaxial and gain structures to overcome the limitations of current approaches and produce the next generation of semiconductor laser technology.

This meeting will be held at the CUI level. Attendance is restricted by registration and to U.S. Persons (U.S. Citizens and Permanent Residents) only.

All attendees will be required to present Government-issued photo identification upon entry to the event. All U.S. Permanent Residents will be required to present their permanent residency card.

REGISTRATION INFORMATION:

Registration is limited by the venue capacity. The maximum number of registrants per organization may be limited. The determination of maximum participants per organization and what constitutes an organization will be made by the DARPA Program Manager. Registrations will be cut off at the registration deadline listed above or once attendance capacity is met, whichever comes first. Early registration is strongly recommended. Any change to the registration deadline will be captured via an amendment to this notice and an update to the registration website. **There will be no on-site registration.**

The Proposers Day is open only to registered potential proposers. The event is closed to the general public and media. An online registration form, preliminary agenda, links to required forms, and meeting details for the Proposers Day can be found at the registration website.

AMPED CUI Materials Request Forms must be filled out by a representative from each organization attending the event, asserting the organization's ability to safeguard CUI. The AMPED CUI Materials Request Forms must be submitted at the time of registration via the registration website.

A completed DARPA Form 104, "DARPA Conference Center Visitor Requirements for Unclassified Meetings" must be filled out by all attendees. For additional information please refer to the Visitor Information page on the DARPA website (http://www.darpa.mil/policy/visitor-information). DARPA Form 104 can be accessed and submitted at https://dtsn.darpa.mil/eform104/visitor/9aa7757f-a017-451c-ae8f-bf484c1dfb65.

Additionally, all meeting registrants who are not U.S. Citizens must complete and submit a DARPA Form 60 "U.S. Permanent Resident and Foreign National Visit Request", which can be submitted via the following link: https://dtsn.darpa.mil/eform60/. Additional questions regarding foreign national visits to DARPA, or its events, may be sent to your DARPA Point of Contact or the Security and Intelligence Directorate International Security staff at SIDInternationalsecurity@darpa.mil.

All forms must be submitted no later than the registration deadline of April 29, 2026, at 5:00 p.m. ET. Failure to complete and submit the required forms prior to the date and time noted above may result in cancellation of existing registration.

TEAMING:
While not required, DARPA is open to teaming amongst the community as appropriate and when strategic to achieve program outcomes. To facilitate the building of teams, Proposers Day registrants may choose to participate in the options below:

1. **Attendee List:** Participant contact information (name, title/position, organization, email address) will be included on an AMPED Proposers Day attendee list distributed to other prospective proposers. Registrants may indicate during registration whether they approve publication of their contact information.

2. **Proposer Profile List**: Interested parties should submit a one-page profile consisting of their contact information (name, title/position, organization, email, telephone number, mailing address, and, if applicable, organization website), a brief unclassified description of their technical competencies, and, if applicable, the expertise desired from other teams/organizations. All profiles must be emailed to AMPED@darpa.mil no later than April 29, 2026, at 5:00 p.m. ET. Following this deadline, the consolidated teaming profiles will be sent via email to the proposers who submitted a profile. Specific content, communications, networking, and team formation are the sole responsibility of the participants. Neither DARPA nor the DoW endorses the information and organizations contained in the consolidated teaming profile document, nor does DARPA or the DoW exercise any responsibility for improper dissemination of the teaming profiles.

3. **Requests Through Email:** Prospective proposers can also request the attendee list and/or the prospective proposer profile list through email at AMPED@darpa.mil. All email requests must be received by June 19, 2026, at 5:00 p.m. ET and follow the same security requirements as Proposers Day attendees (e.g., government photo ID and DARPA Form 60). It is incumbent on email requesters to ensure all provided information is accurate. DARPA will not provide access to secure data transfer capabilities (e.g., DoD SAFE), and DARPA reserves the right to deny requests stemming from its vetting of requesters' security documentation.

POSTERS:
A poster session will be held to facilitate interaction among attendees and provide an opportunity to those who would like to showcase their capabilities and build strong teams to solve the challenge presented in AMPED. There is no specific template for the poster; however, posters should highlight proposer capabilities and teaming requirements and must include presenter's contact information. Posters should not contain any Controlled Unclassified Information (CUI) or classified information and must be suitable for unlimited public release. There is a limit of one poster per organization, and posters should be no larger than 24 x 36 inches. Registrants must indicate their intent to participate in the poster session during registration. Poster presentations will be limited to the first 80 requests due to room constraints, but for any prospective proposer who conveys interest in participating in the poster session after the quantity limit has been reached, that prospective proposer's poster will be distributed to all other confirmed Proposers Day attendees.

BREAKOUT SESSIONS:
One-on-one breakout sessions with the AMPED Program Manager will be available during the Proposers Day. CUI discussion will be allowed. Advance registration is required. 10-minute breakout sessions will be scheduled from 1:00 p.m. to 4:00 p.m. ET. To request a session, please indicate your discussion topic and participating attendees (no more than five) when registering. Please note that all members participating in the breakout sessions must be successfully registered in order to attend. Requests submitted through registration do not guarantee selection for an individual session. All registered attendees chosen to participate in breakout sessions will receive notification by May 6, 2026, containing their selection, scheduled time and location, and detailed guidance.

DARPACONNECT:
Entities who have not worked with DARPA before are encouraged to learn more about DARPAConnect, an initiative established to facilitate collaboration between DARPA and potential performers. The DARPAConnect team offers customized support, resources, and guidance on how to prepare ideas for high-impact conversations with DARPA program managers. Please visit DARPAConnect.us to access a digital hub of online resources, including a curriculum for self-paced learning, personalized support, and in-person and virtual events. To prepare for the Proposers Day, attendees can leverage the learning module on "Making the Most of a DARPA Proposers Day." In addition to the self-paced online materials, the DARPAConnect team is able to schedule one-on-one conversations to discuss specific ideas, questions, and paths to DARPA. Use the contact form at DARPAConnect.us or email the DARPAConnect team directly at darpaconnect@darpa.mil to request assistance.

ADMINISTRATIVE:
All administrative and technical questions should be directed to AMPED@darpa.mil. Please refer to the Special Notice number (DARPA-SN-26-63) in all correspondence.

This Special Notice is issued solely for information and program planning purposes and does not constitute a formal solicitation for proposals or proposal abstracts; any so sent will be disregarded. Attendance is strictly voluntary and shall not obligate, or be construed to require, any attendee to respond to any subsequent Special Notice (SN) or Solicitation related to this topic. DARPA will not provide reimbursement for costs incurred in responding to this Special Notice. Respondents are advised that DARPA is under no obligation to acknowledge receipt of any information received or provide feedback to respondents with respect to any information submitted under this Special Notice.